Chapter 10 — Electromagnetism

Complete Notes · PECTAA Physics 11 · Board-style longs + conceptual shorts

Topic 1 — Force on a current-carrying conductor and magnetic flux

Q.1 Explain the force on a current-carrying conductor in a magnetic field and define magnetic flux and flux linkage.

A conductor of length L carrying current I in a magnetic field B experiences a force F = BIL sin θ, where θ is the angle between the conductor and the field. The direction is given by Fleming's left-hand rule. The tesla is defined from this relation: one tesla gives a force of one newton per metre on a conductor carrying one ampere perpendicular to the field.

Magnetic flux through an area A is Φ = BA cos θ, where θ is the angle between the field and the normal to the area; its unit is the weber. For a coil of N turns, the flux linkage is NΦ. Flux linkage is the quantity that matters in electromagnetic induction.

Important Conceptual Short Questions — Topic 1

1. Why is the force on a conductor maximum when it is perpendicular to the field?

The force is F = BIL sin θ, and sin θ is maximum at 90°. When the conductor is parallel to the field, sin θ = 0 and there is no force.

2. How does Fleming's left-hand rule give the direction of the force?

Hold the thumb, first finger and second finger of the left hand mutually perpendicular. The first finger points along the field, the second along the current, and the thumb gives the direction of the force.

3. Why is flux linkage NΦ used for a coil instead of Φ alone?

Each turn of the coil links the same flux, so the total induced emf depends on the number of turns. The product NΦ gives the total flux linkage of the coil.

studysapiens.com

Topic 2 — Motion of a charged particle and velocity selector

Q.2 Describe the motion of a charged particle in a magnetic field and explain the velocity selector.

A charge q moving with speed v in a magnetic field B experiences a force F = qvB sin θ perpendicular to both v and B. If the particle moves perpendicular to a uniform field, the force is constant in magnitude and always perpendicular to the velocity, so the particle moves in a circle. Equating magnetic force to centripetal force, $qvB = mv^2/r$, gives r = mv/qB.

A velocity selector uses perpendicular electric and magnetic fields. The electric force qE is independent of speed, while the magnetic force qvB depends on speed. Only particles with qE = qvB, that is v = E/B, pass through undeflected.

Important Conceptual Short Questions — Topic 2

1. Why does a magnetic field change the direction of a moving charge but not its speed?

The magnetic force is always perpendicular to the velocity, so it does no work. It changes only the direction of motion, leaving kinetic energy and speed unchanged.

2. How does a velocity selector pass only one speed?

The electric force qE is constant, but the magnetic force qvB increases with speed. They balance only when v = E/B, so particles of that speed go straight while others are deflected.

3. Why does a charged particle move in a circle when it enters a uniform magnetic field perpendicularly?

The magnetic force has constant magnitude and is always perpendicular to the velocity, so it acts as a centripetal force and produces circular motion.

Topic 3 — Induced emf, Faraday's law and Lenz's law

Q.3 Explain motional emf, Faraday's law and Lenz's law.

When a conductor of length L moves with speed v perpendicular to a magnetic field B, an emf $\varepsilon = vBL$ is induced across it. More generally, Faraday's law states that the induced emf is proportional to the rate of change of flux linkage: $\varepsilon = -N\Delta\Phi/\Delta t$. The minus sign represents Lenz's law.

Lenz's law states that the induced current flows in such a direction as to oppose the change producing it. It is a consequence of conservation of energy: if the induced current helped the change, motion would increase without extra work and energy would be created. The induced emf is increased by a stronger field, faster motion, more turns and a larger area.

studysapiens.com

Important Conceptual Short Questions — Topic 3

1. Why is Lenz's law a form of conservation of energy?

The induced current opposes the motion or change that produces it, so external work must be done to keep the change going. This work is converted into electrical energy, so energy is conserved.

2. How does faster motion increase the induced emf?

Faraday's law gives $\varepsilon = -N\Delta\Phi/\Delta t$. Faster motion changes the flux in a shorter time, increasing the rate of change of flux and therefore the induced emf.

3. Why does a magnet fall slowly through a copper tube?

The falling magnet changes the flux through the tube and induces currents. By Lenz's law these currents oppose the motion of the magnet, producing an upward braking force.

Topic 4 — Ferrofluids and seismometers

Q.4 Describe ferrofluids and explain how a seismometer uses electromagnetic induction.

A ferrofluid is a liquid containing nanometre-sized magnetic particles suspended in it. The particles are temporarily magnetized by an external field, so the fluid is attracted to magnets and forms spikes along the field lines. Ferrofluids are used in rotary seals, loudspeaker cooling, targeted drug delivery and damping.

A seismometer detects earthquake motion using a magnet suspended by a spring inside a coil. When the ground moves, the frame and coil move but the magnet tends to stay at rest because of inertia. The relative motion changes the flux through the coil and induces an emf. The induced current is proportional to the velocity of the ground motion, and the signal is amplified and recorded.

Textbook check (pp. 222 and 238): the objectives say the induced emf is proportional to displacement, while p. 238 says velocity. By Faraday's law, velocity is correct.

Important Conceptual Short Questions — Topic 4

1. Why does a ferrofluid form spikes in a magnetic field?

The suspended magnetic particles become magnetized and align along the field. The fluid is pulled into peaks where the magnetic force balances gravity and surface tension, producing spikes.

2. How does inertia help a seismometer detect ground motion?

The suspended magnet tends to remain at rest while the frame and coil move with the ground. This relative motion changes the flux through the coil and induces a measurable emf.

3. Why is the induced emf in a seismometer proportional to velocity rather than displacement?

Faraday's law depends on the rate of change of flux. The rate at which the magnet moves relative to the coil is a velocity, so the induced emf is proportional to velocity.

studysapiens.com